DRONE MOTOR CONTROL / ENDURANCE AND PAYLOAD

Payload or flight time?

Chiralight makes no compromises.

3 kg quad, 10-inch props, 640KV motor and 6S 4 Ah. 1.5:1 TWR (thrust-to-weight ratio) equivalent flight load; separate 3:1 TWR maneuver reserve for gust recovery and obstacle avoidance.

Design freedom: lower cell counts, lower-KV motors, higher minimum RPM and more maneuvering reserve.

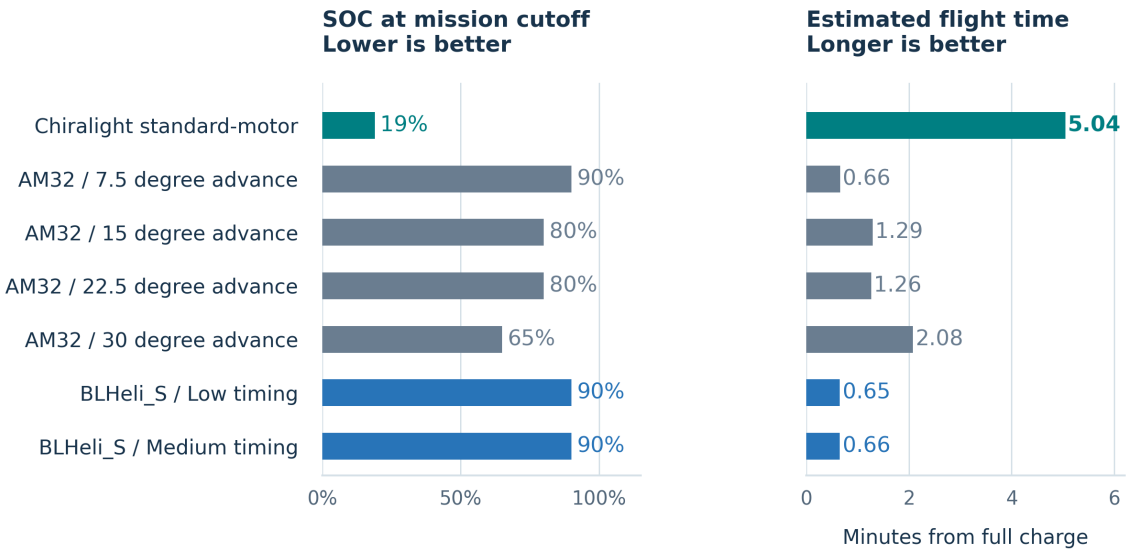
5.0 min
Estimated flight time
Chiralight standard-motor

+143%
Flight time versus the
best tested alternative

+174 g
Potential payload by
choosing 6S over 8S

3:1 TWR MANEUVER RESERVE FOR GUST RECOVERY AND OBSTACLE AVOIDANCE

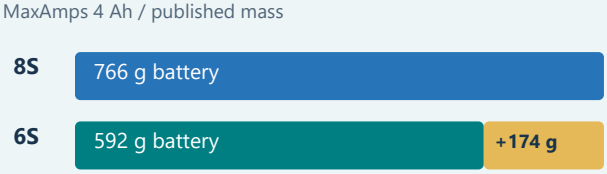
640KV / 6S | 1.5:1 TWR flight load | 3:1 TWR maneuver reserve



AM32 and BLHeli_S show every tested timing setting. The headline compares against **AM32 / 30 degree advance**, the longest-flying tested alternative in this configuration.

6S FOR PAYLOAD. 8S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **174 g for payload** at the same 3 kg all-up mass. With Chiralight standard-motor, 8S estimates 7.91 minutes and adds 174 g.



Matching energy with a lower-Ah 8S pack can leave insufficient power for the 3:1 TWR reserve.

Flight rule: integrate 1.5:1 TWR flight power. Stop at lowest tested SOC meeting 3:1 TWR maneuver reserve for gust recovery and obstacle avoidance, with 3.0 V/cell **MINIMUM** under both loads. Reserve is not continuous. The recommended 3.5 V/cell landing policy is separate.

Model: compiled controllers; nonlinear B/H and table-informed Gemfan 1050 fixture. Synthetic OCV, isothermal 5 milliohms/cell, four motors + 10 W; FOC 1 A injection, MTPA/overmodulation, field weakening off. Equivalent flight estimate, not measured aircraft endurance. Reserve: 125 A mean / 136 A peak against 128 A continuous; no burst rating assumed.

Control comparisons: flight-load efficiency and voltage headroom (p2); dynamic response (p3); representative electronics (p4).

Pack sources: MaxAmps 4 Ah 6S, 592 g / 8S, 766 g. **Verifiable simulation results.** Contact Chiralight Systems. Evaluate for your aircraft. Exact performance may vary.

VOLTAGE AND EFFICIENCY

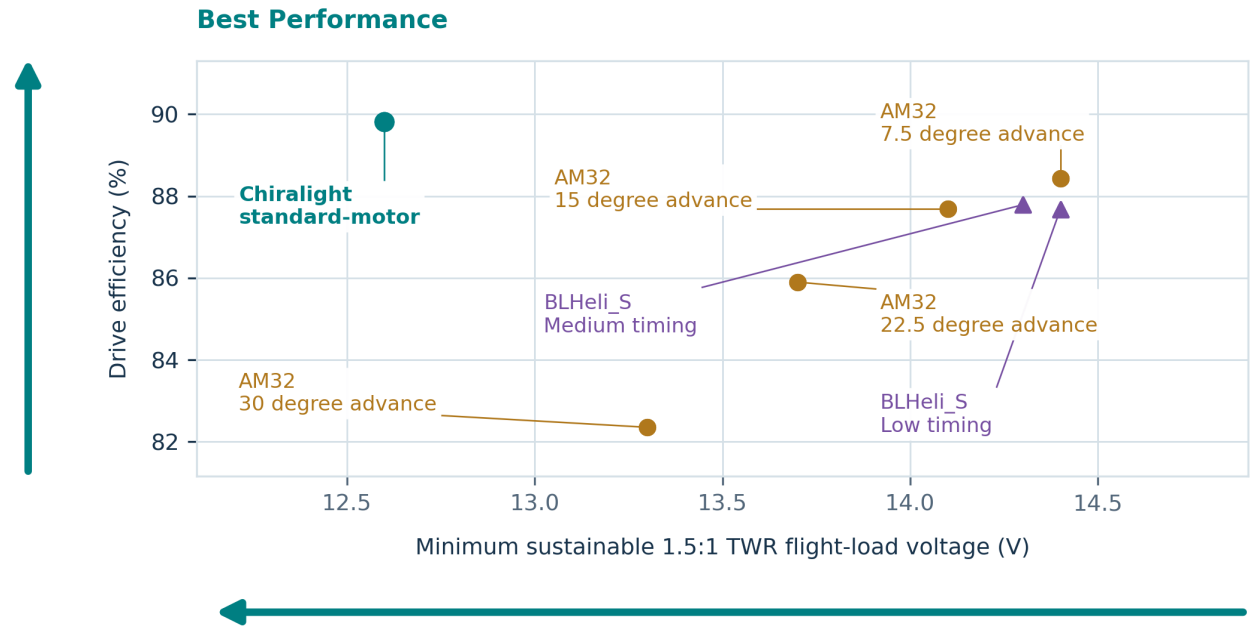
More efficiency. More voltage headroom.

Measured 640KV motor electrical parameters, nonlinear B/H and the same table-informed 10-inch propeller load. Actual compiled controllers; FOC field weakening off, MTPA and overmodulation on.

1.5:1 TWR EQUIVALENT FLIGHT LOAD - 1,125 GF PER ROTOR

Controller / timing	Drive efficiency	Battery to shafts	Minimum flight voltage
Chiralight standard-motor FOC	**89.81%**	85.03%	12.60 V
AM32 / 7.5 degree advance	**88.43%**	83.67%	14.40 V
AM32 / 15 degree advance	**87.69%**	82.93%	14.10 V
AM32 / 22.5 degree advance	**85.90%**	81.16%	13.70 V
AM32 / 30 degree advance	**82.36%**	77.67%	13.30 V
BLHeli_S / Low timing	**87.68%**	82.88%	14.40 V
BLHeli_S / Medium timing	**87.79%**	82.95%	14.30 V

Efficiency at full-charge 6S / 4 Ah, four motors and 10 W auxiliaries; 5 milliohms/cell. Minimum voltage is a separate regulated-bus test at the identical flight load.



Use voltage headroom where the aircraft needs it. Chiralight standard-motor sustains the flight load at **12.60 V**; the lowest tested six-step requirement is **13.30 V**. The chart shows the efficiency cost of each timing setting.

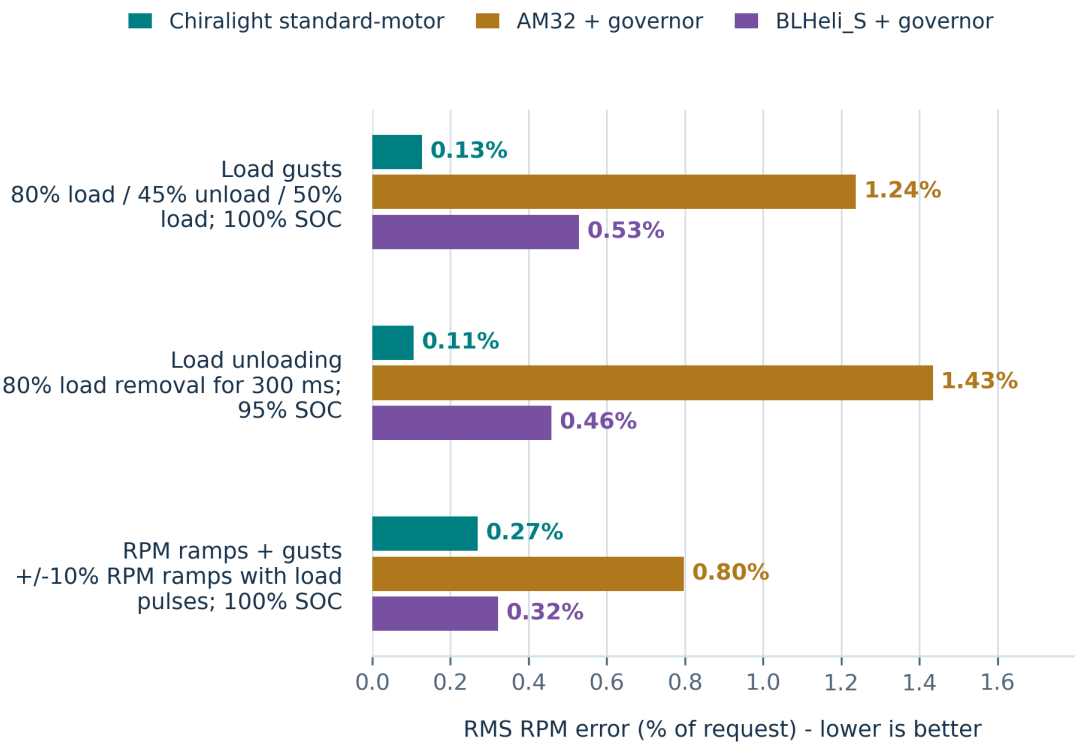
Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.

DYNAMIC RESPONSE AND STARTUP

Hold RPM through changing loads.

Same measured 640KV motor parameters, nonlinear B/H, table-informed propeller and 6S / 4 Ah battery at 5 milliohms/cell. Baseline: 1.5:1 TWR. Each controller uses its sensorless speed estimate; rotor truth only scores the result. Operating SOC is specified for each test.

MATCHED LOAD CHANGES; LOWER RPM ERROR IS BETTER



Separate speed-step training selected the loop settings: Chiralight standard-motor 80 rad/s; AM32 20 rad/s (integral gain x0.5); BLHeli_S 40 rad/s. Gains were frozen before these tests. AM32 and BLHeli_S use external governors driven by native commutation periods.

NATIVE STARTUP AND PROTECTION DIAGNOSTICS

Chiralight standard-motor

3/3 reached control. 3/3 completed; 0 phase trips.

AM32

3/3 reached control. 3/3 completed; 0 phase trips. 1 native startup recovery.

BLHeli_S

3/3 reached control. 3/3 completed; 0 phase trips.

The cards report three selected-gain cold starts per controller. A common 120 A phase guard supplies bench protection. The unloading comparison uses 95% SOC to accept regenerated energy. The separate full-SOC stress diagnostic records energy the held-full battery could not accept.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.

BOARD TIMING AND SENSING

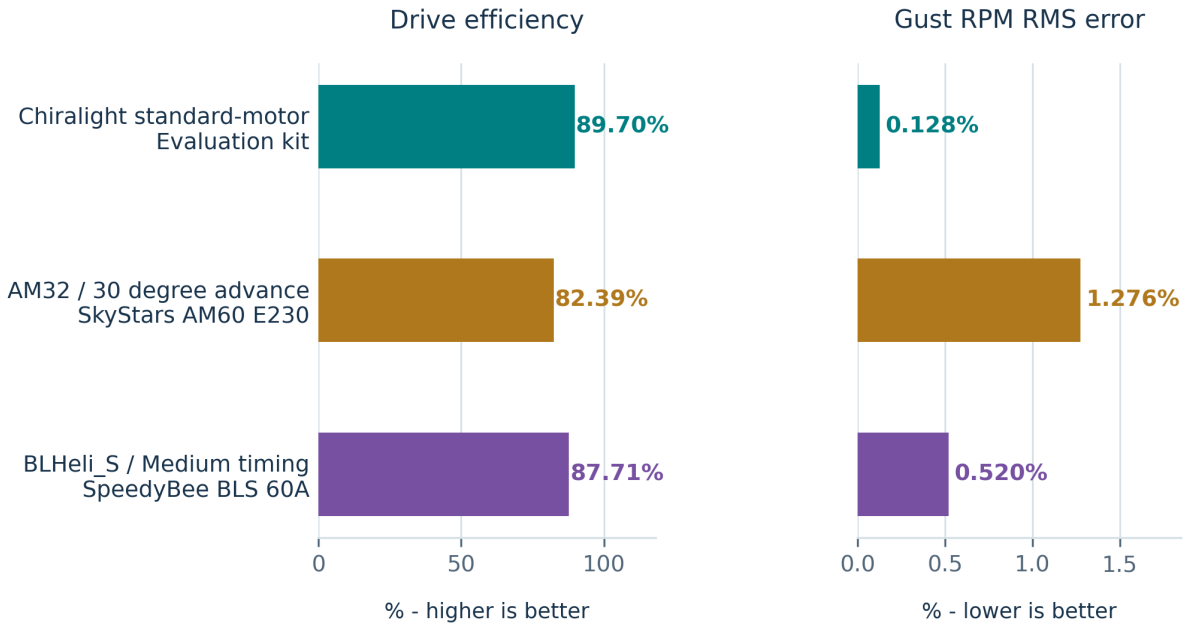
The same motor. Real inverter timing.

Chiralight evaluation-kit sensing and 104 ns blanking, alongside documented representative AM32 and BLHeli_S board profiles. The corrected motor, nonlinear B/H and propeller model remain the same.

SAME FLIGHT LOAD - BOARD-SPECIFIC ELECTRONIC BEHAVIOR

Controller / electronics	DC-to-shaft	Copper loss	Phase RMS
Chiralight standard-motor Chiralight evaluation kit	**89.70%**	21.48 W	11.97 A
AM32 / 30 degree advance SkyStars AM60 E230 profile	**82.39%**	38.29 W	15.98 A
BLHeli_S / Medium timing SpeedyBee BLS60A / J_H_40 profile	**87.71%**	25.58 W	13.06 A

Full-charge 6S / 4 Ah, 5 milliohms/cell, four motors and 10 W auxiliaries. Efficiency: settled 1.5:1 TWR flight load. Gust error: the frozen speed-loop settings from page 3. AM32 uses 30 degree advance, selected for voltage headroom.



TIMING AND MEASUREMENT PROFILES

Chiralight standard-motor: 104.0 ns blanking; 52.08 kHz carrier / 104.17 kHz API sampling; 13 ENOB across +/-250 A; 16-bit converter

AM32 / 30 degree advance: 416.7 ns blanking; 24.00 kHz carrier; native 12-bit DC bus-current ADC, 0-165 A

BLHeli_S / Medium timing: 816.3 ns blanking; 23.93 kHz carrier; no firmware current feedback

Modeled drive efficiency includes winding and diode losses. Unspecified device conduction, switching-energy and magnetic losses are omitted.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.
Evaluate for your aircraft. Exact performance may vary.